

Companion XI

Dark Energy as Emergent Relaxation Dynamics in Relaxation-Driven Cyclic Cosmology (RDCC)

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Abstract

In the standard Λ CDM model, cosmic acceleration is attributed to a dark-energy component or a cosmological constant. In the Relaxation-Driven Cyclic Cosmology (RDCC) framework, acceleration arises instead as an emergent phenomenon of inter-sector relaxation on the bipartite Hilbert space $\mathcal{H}_+ \otimes \mathcal{H}_-$. Building on the microscopic master equation derived in Companion X, we show that the relaxation rate $\Gamma(a) \propto H(a)$ naturally induces an effective acceleration term in the coarse-grained Friedmann dynamics.

The freeze-out of the RDCC relaxation parameter $\alpha(a)$ at late times produces an asymptotic expansion regime that mimics dark energy without introducing a new fundamental energy component. We derive the modified Friedmann equation implied by the RDCC relaxation dynamics, identify the conditions under which the effective acceleration matches observations, and show that the “Why now?” problem is resolved as a structural consequence of the Hubble-scale memory time.

This Companion provides the first RDCC-consistent reinterpretation of dark energy as a dynamical, emergent effect of bipartite decoherence rather than a fundamental energy density.

1 Introduction

The accelerated expansion of the Universe is one of the central empirical discoveries of modern cosmology. In the standard Λ CDM framework, this acceleration is attributed to a dark-energy component with equation of state $w \simeq -1$, often modeled as a cosmological constant. While phenomenologically successful, this interpretation raises several conceptual challenges: the fine-tuning of Λ , the “Why now?” coincidence problem, and the absence of a microscopic explanation for the origin of dark energy.

Relaxation-Driven Cyclic Cosmology (RDCC) offers a structurally different perspective. RDCC describes the Universe as a globally CPT-symmetric quantum state defined on the bipartite Hilbert space $\mathcal{H}_+ \otimes \mathcal{H}_-$, with classical spacetime emerging only after sectoral projection. A central dynamical ingredient of RDCC is the inter-sector relaxation process governed by the dimension-six interaction

$$\mathcal{L}_{\text{int}} = \frac{1}{M^2} T_{\mu\nu}^{(+)} T^{(-)\mu\nu},$$

which induces decoherence between the two sectors. Companion X established that the corresponding relaxation rate satisfies

$$\Gamma(a) \propto H(a),$$

as a structural consequence of stress-tensor correlators evaluated over a Hubble-scale memory time.

This proportionality suggests that cosmic acceleration may arise not from a new energy component, but from the relaxation dynamics themselves. As the Universe expands and $H(a)$ decreases, the relaxation rate weakens and the coherence parameter $\alpha(a)$ approaches a constant value. The freeze-out of $\alpha(a)$ modifies the coarse-grained Friedmann dynamics and can generate an effective acceleration term that mimics dark energy without introducing a fundamental cosmological constant.

The purpose of this Companion is to develop this mechanism systematically. We derive the modified Friedmann equation implied by RDCC relaxation, identify the conditions under which the resulting expansion reproduces the observed late-time acceleration, and show that the “Why now?” problem is resolved as a structural consequence of the Hubble-scale memory time inherent in the relaxation process. This establishes a unified interpretation of dark energy as emergent relaxation dynamics within the RDCC framework.

2 RDCC Relaxation Dynamics

The RDCC framework is built on the observation that the global state of the Universe lives on the bipartite Hilbert space $\mathcal{H}_+ \otimes \mathcal{H}_-$, with the two sectors related by global CPT symmetry. Classical spacetime and causal structure emerge only after projection onto one sector, while the global state remains fully quantum and CPT-symmetric.

A central dynamical ingredient of RDCC is the decay of inter-sector coherence. The off-diagonal components of the global density matrix,

$$\rho_{+-}, \quad \rho_{-+},$$

encode the degree of correlation between the sectors. Their decay is governed by the relaxation equation

$$\dot{\alpha} = -\Gamma\alpha,$$

where $\alpha(a)$ quantifies the remaining coherence and $\Gamma(a)$ is the relaxation rate.

Companion VI introduced this equation phenomenologically and showed that it implies a multiplicative evolution of $\alpha(a)$ with a natural late-time freeze-out once $\Gamma(a)$ becomes small compared to the expansion rate.

Companion X provided the microscopic foundation for this behaviour. Starting from the dimension-six interaction $\mathcal{L}_{\text{int}} = \frac{1}{M^2} T_{\mu\nu}^{(+)} T^{(-)\mu\nu}$, a Born–Markov master equation was derived for ρ_{global} in an expanding FRW background. The resulting Lindblad evolution yields the universal scaling

$$\Gamma(a) \propto H(a),$$

as a consequence of relativistic stress-tensor correlators evaluated over a Hubble-scale memory window.

This proportionality ties the relaxation dynamics directly to the expansion rate. As the Universe expands and $H(a)$ decreases, the relaxation rate weakens proportionally, eventually causing $\alpha(a)$ to freeze out. This freeze-out marks the transition to a regime in which inter-sector decoherence becomes ineffective and the emergent classical spacetime structure stabilizes.

In the remainder of this Companion, we investigate how this relaxation process modifies the coarse-grained Friedmann equations and how the freeze-out of $\alpha(a)$ can generate an effective acceleration term that mimics dark energy without introducing a new fundamental energy component.

3 Coarse-Grained Friedmann Dynamics

To determine how relaxation affects cosmic expansion, we derive the coarse-grained Friedmann dynamics implied by the RDCC relaxation process. The key observation is that inter-sector decoherence modifies the effective stress-energy content of the emergent classical spacetime. Rather than introducing a new energy component, the relaxation dynamics alter the relation between the expansion rate and the effective energy density.

3.1 Effective Energy Balance

In standard FRW cosmology, the expansion rate satisfies

$$H^2(a) = \frac{8\pi G}{3} \rho(a), \quad (1)$$

where $\rho(a)$ is the total matter–radiation energy density. In RDCC, the effective energy density relevant for the emergent classical spacetime is modified by the relaxation dynamics. We therefore define

$$H^2(a) = \frac{8\pi G}{3} \rho_{\text{eff}}(a), \quad (2)$$

where $\rho_{\text{eff}}(a)$ incorporates the influence of the coherence parameter $\alpha(a)$.

3.2 Relaxation-Induced Correction

Companion X established the universal scaling

$$\Gamma(a) \propto H(a),$$

and the coherence evolution

$$\dot{\alpha}(a) = -\Gamma(a) \alpha(a).$$

As the Universe expands and $H(a)$ decreases, the relaxation rate weakens and $\alpha(a)$ approaches a constant value, marking the freeze-out of inter-sector decoherence.

Tracing over the opposite sector suppresses the off-diagonal components of ρ_{global} , effectively reducing the contribution of inter-sector correlations to the emergent stress-energy tensor. This yields a correction of the form

$$\rho_{\text{eff}}(a) = \rho(a) + \Delta\rho_{\text{rel}}(a), \quad (3)$$

where $\Delta\rho_{\text{rel}}(a)$ encodes the relaxation-induced modification.

3.3 Form of the Relaxation Correction

The structure of the Lindblad kernel derived in Companion X implies

$$\Delta\rho_{\text{rel}}(a) \propto \alpha(a) \Gamma(a). \quad (4)$$

Using $\Gamma(a) \propto H(a)$, we obtain the scaling

$$\Delta\rho_{\text{rel}}(a) \propto \alpha(a) H(a). \quad (5)$$

This term acts as an effective acceleration source in the coarse-grained Friedmann equation. Crucially, it does not correspond to a new fundamental energy density; it arises from the dynamical suppression of inter-sector coherence.

In the next section, we show that this relaxation-induced correction can mimic the observational effects of dark energy and naturally produces an asymptotic expansion regime once $\alpha(a)$ freezes out.

4 Emergent Dark-Energy Behaviour

The relaxation-induced correction $\Delta\rho_{\text{rel}}(a)$ modifies the coarse-grained Friedmann dynamics in a way that can mimic the observational effects of dark energy. We now analyse the conditions under which the RDCC relaxation term behaves like an effective dark-energy component and drives accelerated expansion.

4.1 Effective Dark-Energy Component

We rewrite the coarse-grained Friedmann equation as

$$H^2(a) = \frac{8\pi G}{3} [\rho(a) + \rho_{\text{DE}}(a)], \quad (6)$$

where we identify

$$\rho_{\text{DE}}(a) = \Delta\rho_{\text{rel}}(a) \propto \alpha(a) H(a). \quad (7)$$

The effective dark-energy density is therefore not a fundamental fluid but a dynamical quantity tied to the relaxation of inter-sector coherence.

To characterise its behaviour, we introduce an effective equation-of-state parameter $w_{\text{DE}}(a)$ via the continuity equation

$$\frac{d\rho_{\text{DE}}}{da} = -\frac{3}{a} [1 + w_{\text{DE}}(a)] \rho_{\text{DE}}(a). \quad (8)$$

4.2 Acceleration Condition

Cosmic acceleration occurs when

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3} [\rho_{\text{tot}}(a) + 3p_{\text{tot}}(a)] > 0. \quad (9)$$

In the presence of an effective dark-energy component, this requires

$$w_{\text{DE}}(a) < -\frac{1}{3}.$$

The relaxation-induced term

$$\rho_{\text{DE}}(a) \propto \alpha(a) H(a)$$

inherits its time dependence from both the expansion rate and the evolution of $\alpha(a)$. During epochs in which $\Gamma(a)$ is large, $\alpha(a)$ evolves significantly and $\rho_{\text{DE}}(a)$ tracks the expansion. As $H(a)$ decreases and the relaxation rate weakens, $\alpha(a)$ approaches a constant value and $\rho_{\text{DE}}(a)$ enters an asymptotic regime.

4.3 Asymptotic Regime and Dark-Energy Mimicry

In the late-time regime where $\alpha(a)$ has frozen out, we approximate

$$\alpha(a) \simeq \alpha_0 = \text{const.}$$

The effective dark-energy density then scales as

$$\rho_{\text{DE}}(a) \propto H(a).$$

In a regime where the expansion rate varies slowly, this behaviour mimics an approximately constant dark-energy density with $w_{\text{DE}} \simeq -1$, reproducing the key phenomenological feature of Λ CDM without introducing a fundamental cosmological constant.

The emergent dark-energy behaviour in RDCC is therefore a consequence of the interplay between inter-sector relaxation, the Hubble-scale memory time, and the freeze-out of $\alpha(a)$.

5 The “Why now?” Problem in RDCC

A longstanding challenge in dark-energy cosmology is the “Why now?” problem: why does the onset of cosmic acceleration occur precisely when matter and dark-energy densities are of comparable magnitude? In Λ CDM this coincidence appears accidental, as the cosmological constant is time-independent while matter redshifts as a^{-3} .

In RDCC, the timing of acceleration is not a coincidence but a structural consequence of the relaxation dynamics. The key observation is the universal scaling

$$\Gamma(a) \propto H(a),$$

which implies that the ratio of relaxation and expansion timescales is approximately constant:

$$\frac{\Gamma(a)}{H(a)} = \text{const.}$$

5.1 Crossing of Relaxation and Expansion Timescales

The coherence parameter evolves according to

$$\dot{\alpha}(a) = -\Gamma(a) \alpha(a).$$

As long as $\Gamma(a)$ is large compared to the rate of change of $H(a)$, the relaxation process is efficient and $\alpha(a)$ evolves significantly. As the Universe expands and $H(a)$ decreases, the relaxation rate weakens proportionally, eventually leading to a regime in which

$$\Gamma(a) \ll H(a).$$

In this regime, relaxation becomes ineffective and $\alpha(a)$ freezes out.

The onset of freeze-out therefore occurs when the relaxation and expansion timescales cross:

$$\Gamma(a_{\text{fr}}) \simeq H(a)_{\text{fr}}.$$

This crossing is not fine-tuned; it follows directly from the proportionality $\Gamma(a) \propto H(a)$.

5.2 Natural Timing of Acceleration

The effective dark-energy density scales as

$$\rho_{\text{DE}}(a) \propto \alpha(a) H(a).$$

Before freeze-out, $\alpha(a)$ evolves rapidly and $\rho_{\text{DE}}(a)$ tracks the expansion rate. After freeze-out, $\alpha(a)$ becomes approximately constant and $\rho_{\text{DE}}(a)$ enters an asymptotic regime that can mimic a cosmological constant.

The onset of acceleration therefore coincides with the freeze-out of $\alpha(a)$:

$$a_{\text{acc}} \simeq a_{\text{fr}}.$$

This provides a natural explanation for the timing of cosmic acceleration: it occurs precisely when the relaxation dynamics become inefficient.

5.3 Resolution of the Coincidence Problem

In RDCC, the “Why now?” problem is resolved without introducing a new energy component or fine-tuning parameters. The coincidence between the onset of acceleration and the present epoch arises because both are governed by the same dynamical mechanism:

Acceleration begins when relaxation freezes out.

This structural relation replaces the coincidence with a dynamical explanation rooted in the Hubble-scale memory time of the relaxation kernel.

6 Predictions and Observational Signatures

Interpreting dark energy as emergent relaxation dynamics leads to several distinctive observational signatures. While the effective late-time behaviour of the RDCC relaxation term can mimic a cosmological constant, the underlying dynamics differ from Λ CDM in ways that may be testable.

6.1 Deviation from a Constant Equation of State

In Λ CDM, the dark-energy equation of state is strictly constant, $w_\Lambda = -1$. In RDCC, the effective dark-energy density

$$\rho_{\text{DE}}(a) \propto \alpha(a) H(a)$$

implies a dynamical equation-of-state parameter

$$w_{\text{DE}}(a) = -1 + \frac{d \ln H}{d \ln a}.$$

During the freeze-out of $\alpha(a)$, small deviations from $w = -1$ may occur. These deviations are expected to be mild but could be detectable with high-precision late-time cosmological probes.

6.2 Late-Time Transition in the Expansion Rate

The freeze-out of $\alpha(a)$ induces a transition in the effective expansion dynamics. Before freeze-out, the relaxation term tracks the Hubble rate; after freeze-out, it approaches an asymptotic regime. This leads to a characteristic feature:

$$\frac{dH}{da} \text{ changes slope near } a \simeq a_{\text{fr}}.$$

Such a transition could manifest as a mild redshift-dependent deviation from the Λ CDM expansion history, potentially observable in:

- baryon acoustic oscillations (BAO),
- Type Ia supernovae,
- redshift-drift measurements,
- standard-siren observations.

6.3 Tensor-Sector Implications

The relaxation dynamics are tied to the same inter-sector structure that generates log-periodic modulation in the primordial tensor spectrum, as discussed in Companions II–V. If the effective dark-energy behaviour arises from the freeze-out of $\alpha(a)$, then the timing of this freeze-out may correlate with the amplitude or phase of tensor-sector modulation.

This yields a potential cross-correlation signature:

$$\text{Late-time acceleration} \iff \text{specific tensor-modulation phase.}$$

6.4 Absence of a True Cosmological Constant

A key observational consequence of RDCC is the absence of a fundamental cosmological constant. Instead, the effective dark-energy density arises dynamically from the relaxation process. This implies that

$$\lim_{a \rightarrow \infty} \rho_{\text{DE}}(a) \quad \text{need not be strictly constant.}$$

Future measurements of the late-time expansion rate, especially those sensitive to the asymptotic behaviour of $H(a)$, may therefore distinguish RDCC from models with a true cosmological constant.

6.5 Summary of Observational Signatures

The RDCC interpretation of dark energy predicts:

- mild deviations from $w = -1$ during the freeze-out of $\alpha(a)$,
- a characteristic transition in the slope of $H(a)$ near a_{fr} ,
- possible correlations between late-time acceleration and tensor-sector modulation,
- absence of a strictly constant asymptotic dark-energy density.

These signatures provide potential avenues for testing RDCC and distinguishing emergent relaxation dynamics from a fundamental cosmological constant.

7 Discussion and Outlook

We have developed a reinterpretation of dark energy within the Relaxation-Driven Cyclic Cosmology framework. Building on the microscopic relaxation dynamics derived in Companion X, we have shown that the suppression of inter-sector coherence induces an effective modification of the coarse-grained Friedmann equations. This modification behaves like a dynamical dark-energy component, even though no new fundamental energy density is introduced.

The key structural result is the universal scaling

$$\Gamma(a) \propto H(a),$$

which ties the relaxation dynamics directly to the expansion rate. As the Universe expands and $H(a)$ decreases, the relaxation becomes inefficient and $\alpha(a)$ freezes out. This freeze-out naturally produces an asymptotic expansion regime that can mimic a cosmological constant, providing a structural explanation for late-time acceleration.

A particularly noteworthy consequence is the resolution of the “Why now?” problem. In RDCC, the onset of acceleration coincides with the freeze-out of $\alpha(a)$, which occurs when the

relaxation and expansion timescales cross. This timing is not fine-tuned but follows directly from the proportionality $\Gamma(a) \propto H(a)$ and the Hubble-scale memory time of the relaxation kernel.

The emergent dark-energy behaviour predicted by RDCC leads to several potential observational signatures, including mild deviations from $w = -1$, a characteristic transition in the slope of $H(a)$ near the freeze-out epoch, and possible correlations with tensor-sector modulation. These signatures provide avenues for distinguishing RDCC from Λ CDM and other dynamical dark-energy models.

Future Directions

The results presented here open several directions for further development:

- **Modified perturbation theory.** A full treatment of scalar, vector, and tensor perturbations in the presence of relaxation-induced corrections may reveal additional observational signatures.
- **Non-Markovian effects.** The derivation in Companion X relies on the Born–Markov approximation. Including non-Markovian corrections may refine the effective dark-energy behaviour, especially near the freeze-out epoch.
- **Numerical implementation.** Incorporating the relaxation correction $\Delta\rho_{\text{rel}}(a)$ into a Boltzmann code would allow for global parameter scans and quantitative comparison with cosmological data.
- **Information-theoretic interpretation.** The connection between decoherence, emergent spacetime, and effective acceleration suggests a deeper information-theoretic structure underlying cosmic expansion.

RDCC provides a unified and structurally motivated reinterpretation of dark energy as emergent relaxation dynamics. This perspective integrates cosmic acceleration into the same framework that governs inter-sector decoherence, tensor-sector modulation, and the emergence of classical spacetime.

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